

SOM-based Contamination Decay Regime Discovery

Methodology pipeline: from JAEA EMDb acquisition to four discrete regime labels

DATA PREPARATION · Section 2.1

EMDB raw archive

14.5 M records
140,755 mesh cells

FY 2011-2024

Four filters applied

≤ 80 km from FDNPP

dose > 0.10 μSv/h

≥ 4 measurement dates

single survey type

138,768 cells retained

98.6% of acquired cells

modeling input

MODELING PIPELINE · Sections 2.2-2.4

1

Feature Matrix

Eight temporal decay features

• decay ratio · log-linear slope & R²

• early- & late-phase slopes

• early/late slope ratio

• residual CV · uptick rate

138,768 × 8 · robust-IQR z-scored

2

SOM Training

Hexagonal Kohonen grid

• 5×5 & 6×6 hex grids

• 5 seeds × 2,000 iterations

• α 0.05→0.01 · Euclidean · Gaussian

10 candidate models

3

Model Selection

Map quality + seed stability

• QE 1.01 · TE 0.29 (reported, not gated)

• seed-ARI 0.77 (substantial)

• 5×5 grid · lowest-QE seed 456

selected: 5×5 / seed 456

4

Hierarchical Clustering

Ward's D2 on codebook

• 25 codebook vectors

• silhouette flat k=3-6 · gap → k=1

• k=4: balance + interpretability

k = 4

OUTPUT · four decay regimes

per-cell labels assigned to all 138,768 mesh cells

R1

Slow Decay

share20.71%

t½10.45 y

Dist.61.53 km

Dose0.226 μSv/h

R2

Standard Decay

share59.81%

t½5.17 y

Dist.54.33 km

Dose0.778 μSv/h

R3

Near-Field Hotspot

share14.30%

t½3.80 y

Dist.28.45 km

Dose7.565 μSv/h

R4

Irregular Decay

share5.19%

t½8.91 y

Dist.53.49 km

Dose0.476 μSv/h